Homework 5

Solve the following problems justifying all your answers. You should not use a graph calculator to justify any answer.

1. Compute the derivatives of the following functions:

(a) $T(z) = 2^z \log_2 z$,

(b) $y(x) = \ln\left(x + \sqrt{x^2 - 1}\right)$.

2. (a) Find all points on the graph of the function $f(x) = -2\sin(x) + \sin^2(x)$ at which the tangent line is horizontal.

(b) At what point on the curve $y = \sqrt{1 + 2x}$ is the tangent line perpendicular to the line $6x + 2y = 1$?

(c) Find all points (x, y) on the graph of $f(x) = 3x^2 - 4x$ with tangent lines parallel to the line $y = 8x + 5$.

3. Prove the power rule for rational numbers by implicit differentiation; i.e. prove that $\left(x^{p/q}\right)' = \dfrac{p}{q} x^{p/q-1}$ for $\frac{p}{q} \in \mathbb{Q}$ by implicit differentiation.

4. Prove that $\lim\limits_{n\to\infty} \left(1 + \dfrac{x}{n}\right)^n = e^x$.